

Summary

$$T_{base}^{tool}(q) = \begin{pmatrix} R(q) & p(q) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R(q) = [r_1 \ r_2 \ r_3]$$

- 3x3 matrix $R(q)$ specifies the *orientation* of the tool, while the 3x1 matrix
- $p(q)$ specifies the *position* of the tool tip.

Tool configuration vector $\in R^6$

$$\begin{pmatrix} p \\ \exp(q_n / \pi) r^3 \end{pmatrix}$$

